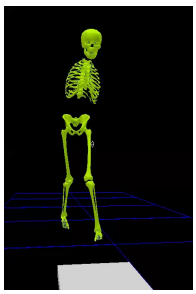
Lecture 11 Pathological gait

SM2501/FSG3084

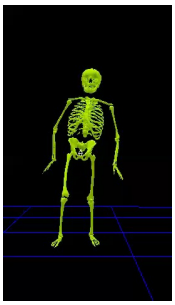




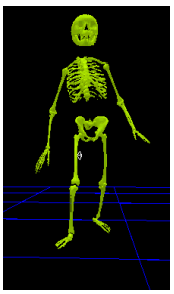
1



Child with cerebral palsy



Child with muscular dystrophy



Child with spina bifida

Why does s/he walk like this?

'Cause' and 'effect'?
What should we treat? How?
What should we expect?
What will happen if we don't?
If we could do it all over again....?

2

Why study gait in pathology?

- One of the most basic functions
  - Efficient, optimal energy conservation
  - "Patients strive to retain this capability even in the presence of severe impairment. As … pathology alter[s] mobility, … patients substitute wherever possible, yield when they must, and accept compensatory reactions…" (Jacquelin Perry 1992)

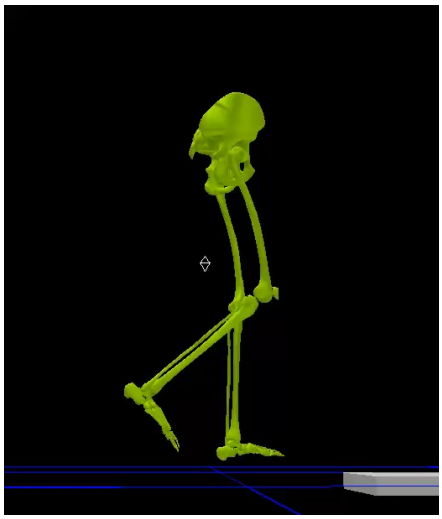
3

Gait deviations and combinations

Deviations, compensatory movements and cross-plane interactions

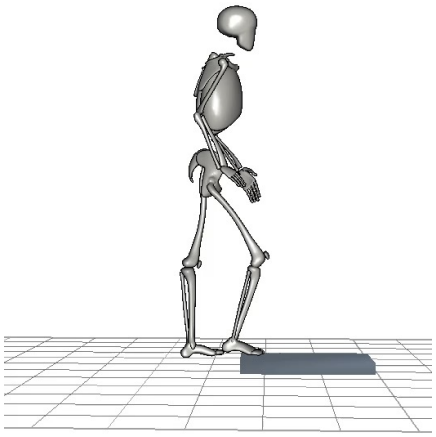
4

Stiff knee/reduced knee flexion in swing
&
Hip abduction in swing (circumduction)



5

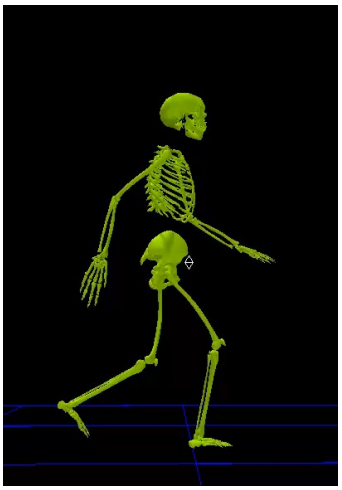
Stiff knee in swing
&
Hip abduction in swing (circumduction)
&
Pelvic elevation in swing (hike)

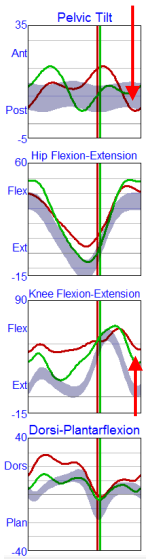


Also excessive plantarflexion-knee extension couple on left

6

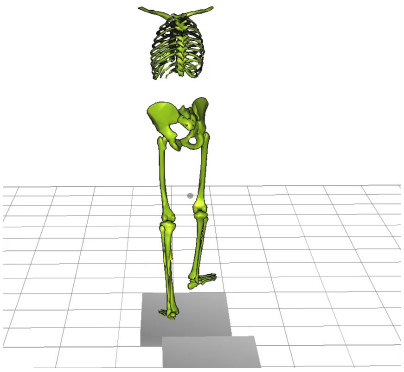
Persistent knee flexion in late swing
&
Posterior pelvic tilt in late swing

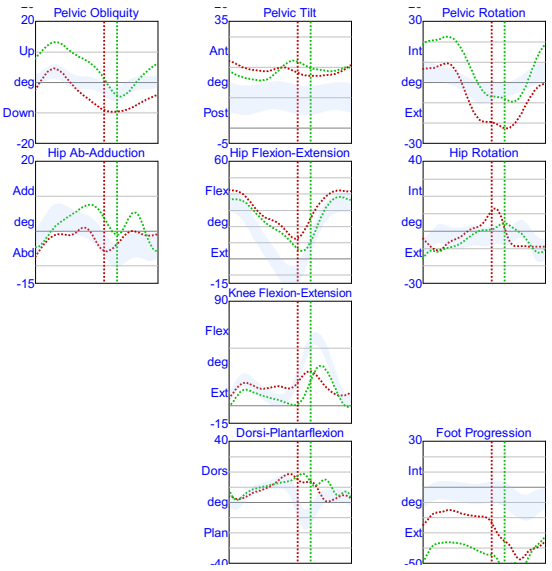




7

Reduced sagittal plane motion
&
Increased transverse plane motion

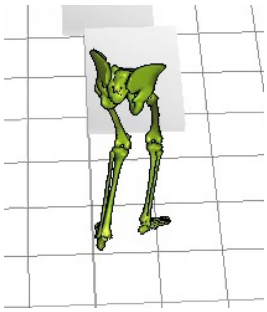


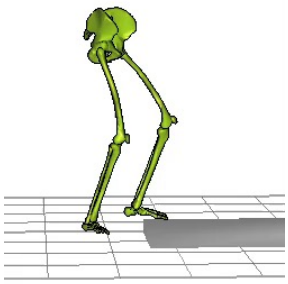


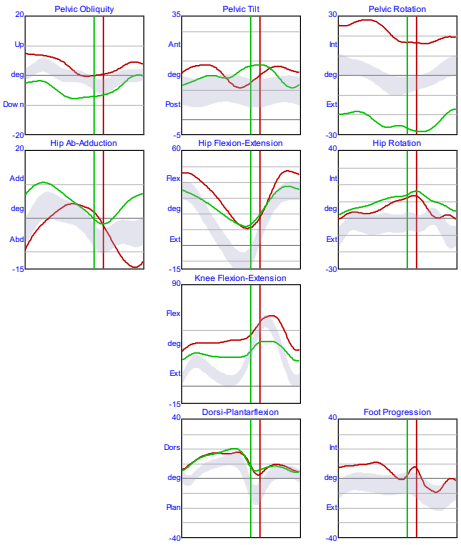
8

Asymmetric pelvis rotation (transverse plane) &

Hip ab-adduction (frontal plane) & reduced sagittal plane movements







9

Gait pathologies and deviations

- Conditions that impair walking differ markedly in primary pathology, yet…
- Fall into 5 functional categories
  - Deformity
  - Muscle weakness
  - Sensory loss
  - Pain
  - Impaired motor control
- 48 single gait deviations

Perry and Burnfield, 2010

10

Neuromuscular origin

11

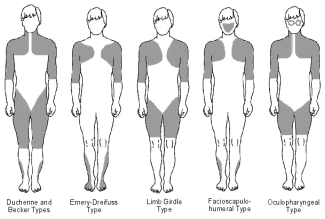
Muscle weakness

- Disorder, diseases, hormonal conditions, electrolyte imbalances, vitamin D deficiency, etc.
- Infections (flu, Lyme, meningitis, COVID, EBV, etc.)
- Neuralgia, anemia, diabetes
- Stroke
- Herniated disk
- Peripheral neuropathy
- Prolonged bed rest
- Alcoholism
- Sleep disorders
- Side effect of corticosteroids, statins, penicillin etc.
- Infections
- Injury
- Aging

12

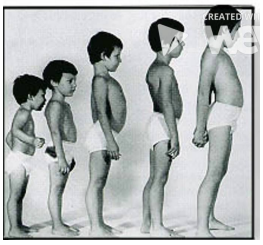
Muscle weakness

- Muscular disease



Main areas of muscle weakness in different types of dystrophy

Muscular dystrophies





Duchenne muscular dystrophy



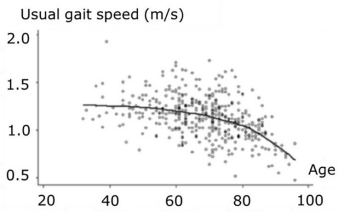


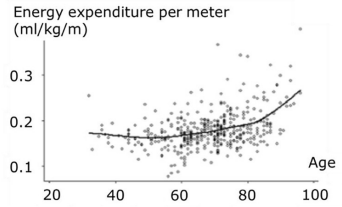
Polio

13

Muscle weakness (sarcopenia) associated with aging

- Velocity typically reduces after age 70.
  - Typicaly unchanged cadence but shorter steps, attributed to weakening calf muscles
  - Gait velocity is a powerful predictor of mortality
- Proportion of double stance time increases
  - Even more so with balance impairment, slippery surfaces, and fear of falling
- Posture changes slightly
  - Greater anterior pelvic tilt and lumbar lordosis
    - Atrributable to weakening abdominal muscles and tight hip flexor muscles
  - Increased external foot progression
    - Attributable to reduced hip internal ROM and/or strategy to increase lateral stability
  - Reduced ankle plantarflexion in pre-swing





Boyer et al., Exper. Geron. 2023

14

Impaired motor control

- Basic impairments:
  - Muscle weakness
  - Impaired selective control
    - Difficulty in controlling timing and intensity of muscle actions
    - Muscle synergies and co-contractions often present
  - Primitive locomotor patterns
    - Instead of voluntary control
    - Mass flexion pattern and mass extension pattern of muscle synergies
  - Spasticity
    - Approx: velocity-induced increase in muscle tone

15

Spinal cord injury

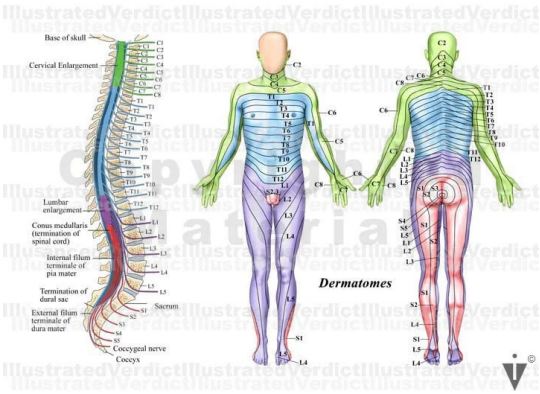
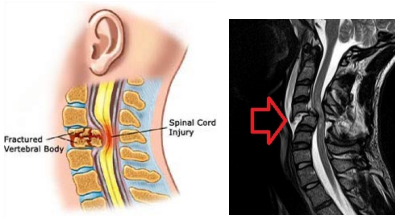
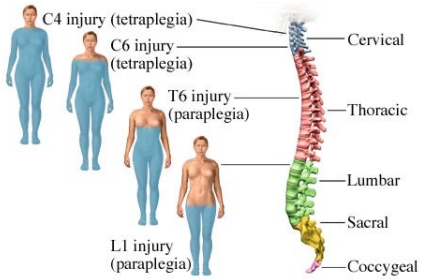


Image: Illustrated Verdict





Classified clinically according to observed motor and sensory function

16

Spina bifida

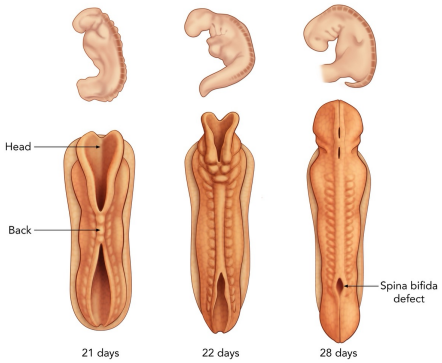




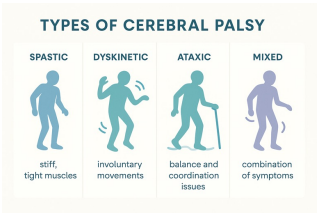
Image: Children's Minnesota

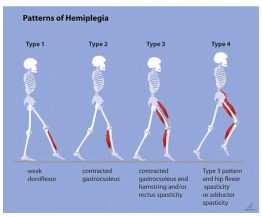
17

Bleeding, lesion or oxygen loss to immature brain (Cerebral palsy)

Typically: combination of muscle weakness, spasticity, and impaired voluntary motor control

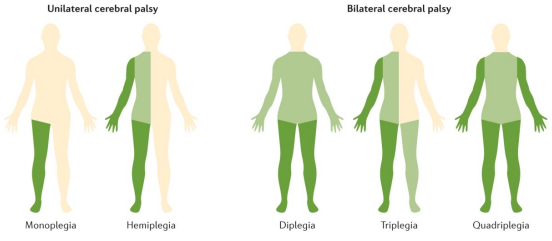
Image: Neomotion

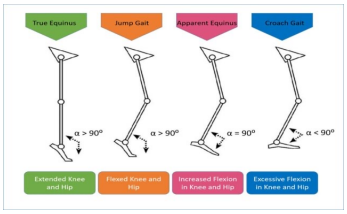




Miller et al (eds). Cerebral Palsy. Springer 2020

Graham et al, 2016



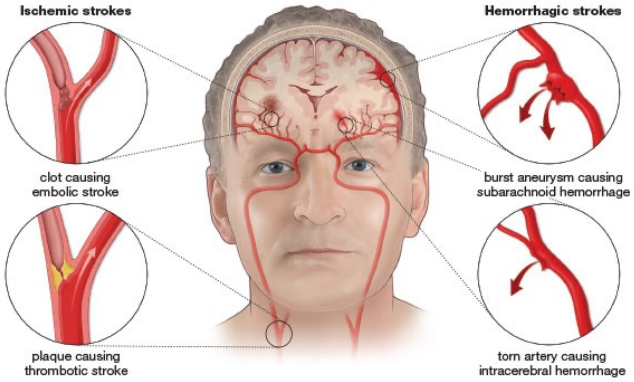


Yelle & Reka, 2024

18

Bleeding, lesion or oxygen loss to mature brain (Stroke)

Typically: combination of muscle weakness, spasticity, impaired voluntary motor control, impaired balance, impaired proprioception



19

Sensory Loss

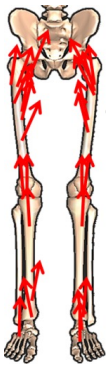
- Impaired tactile sensation
- Impaired proprioception and balance
- Impaired joint proprioception
- Often results in slow, cautious gait

20

Bone and joint deformities

21

Secondary developmental bone and joint deformities from neuromuscular origin



Atypical excitation

-> Atypical tone

-> Atypical forces during growth

-> Joint deformities (soft tissues)

-> Bone deformities

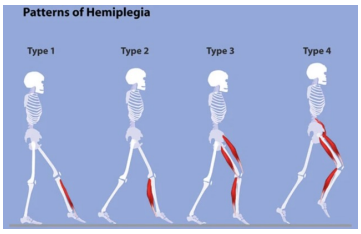
22

Joint deformity, muscles and soft tissues

Joint contracture: reduced joint ROM due to structural changes in fibrous connective tissue components of muscles, ligaments or joint capsule

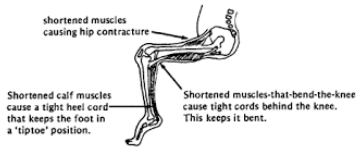
- Dynamic contracture: extension/flexion deficit due to spasticity





Miller et al (eds). Cerebral Palsy. Springer 2020

- Fixed contracture: described as fixed position, or as deficit of opposite movement



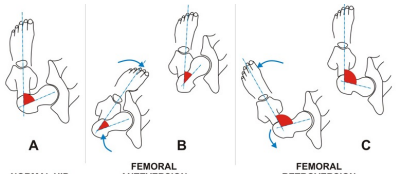


Knee flexion contracture
Knee extension deficit

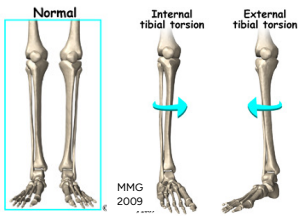
23

Rotational Bone Deformity

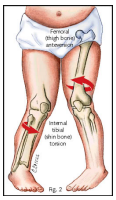
Femoral anteversion or retroversion



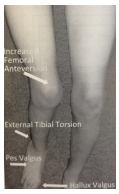
Tibial torsion



Combinations

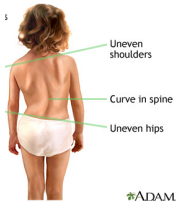


Femoral anteversion
Internal tibial torsion



Femoral anteversion
External tibial torsion

Frontal plane malalignments



Scoliosis



Knee valgus



Knee varus



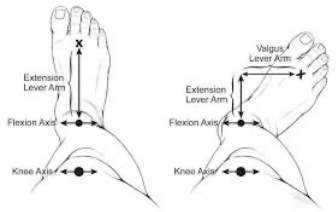
Calcaneus valgus or varus

24

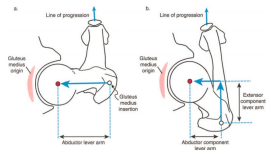
Lever arm dysfunction

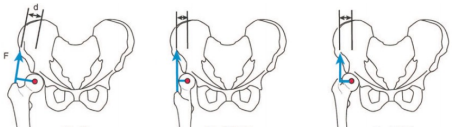
Fixed/bony deformities

Tibia



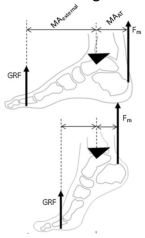
Femur



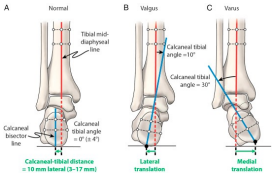


Dynamic deformities

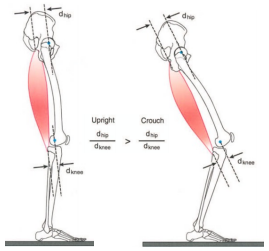
Ankle sagittal



Ankle frontal



Knee sagittal



Paley et al 2002, Kalkman et al 2017

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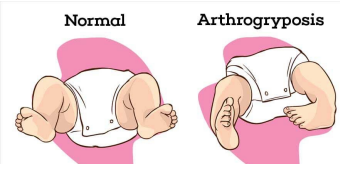
Joint deformity, bone and joint

Joint contracture: reduced joint ROM due to structural changes in fibrous connective tissue components of muscles, ligaments or joint capsule

- Abnormal joint and bone topology

Congenital

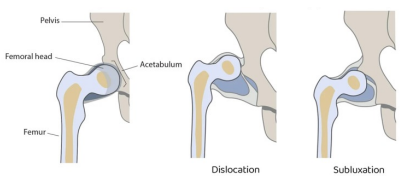




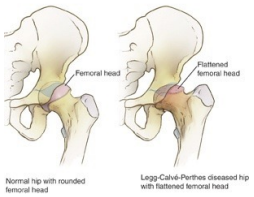
Acquired

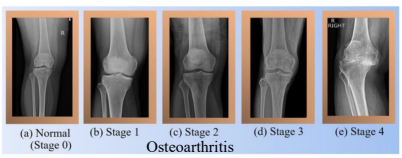


Polichetti et al. *Diagnostics* **2023**



Secondary





26

Pain



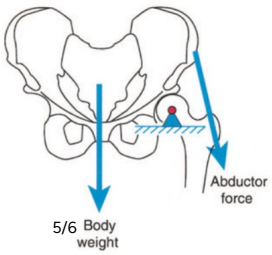
Often results in slow gait, avoidance of a side/joint, etc.

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Group exercise

Part 1: In the figure to the right, assume it loosely depicts a non-disabled person standing on 1 foot. Assume for this exercise that only the abductors are active at the hip, and can be modelled as 1 abductor muscle along this vector line of action.

- Rescale the Abductor muscle force vector length and approximate its magnitude in proportion BW.
- Draw a scaled vector of the hip joint contact force.



Part 2: Now you can change the segment positions.

- Arrange the body segments so the abductor muscle force can (hypothetically) be reduced to 0. A stick figure of a whole body might help.
- Draw a scaled vector of the hip joint contact force. Compare it to the joint contact force vector in the Part 1. Comments?

28